CC 101 – Introduction to Computing

"From Problem to README": Designing a Number-System Converter and Documenting It in Markdown

Answer sheet

Name: Yhamel M. De Guzman	Block: 1
Computer Unit Used (Desktop/Laptop, brand): Laptop, Acer	Date: 20-10-2026

Part A – Analyze the Problem (Problem 1)

Step 1 – Hand conversion (divide by 2, then read the remainders from bottom to top):

Decimal	Division steps	Binary
13	13 ÷ 2 = 6 r 1; 6 ÷ 2 = 3 r 0; 3 ÷ 2 = 1 r 1; 1 ÷ 2 = 0 r 1	1101
25	25 ÷ 2 = 12 r 1; 12 ÷ 2 = 6 r 0; 6 ÷ 2 = 3 r 0; 3 ÷ 2 = 1 r 1; 1 ÷ 2 = 0 r 1	11001

Table 1. Problem Analysis for Problem 1 (Part A)

Item	Your Answer
Restated problem (your own words)	Take a whole number written in decimal (0 or greater) and turn it into binary. Keep dividing the number by 2 and saving the remainders; the remainders, read from last to first, are the binary digits that get displayed.
Input	n – a whole decimal number, n ≥ 0 (entered by the user)
Output	The binary form of n, shown as a string of 0s and 1s (example: 13 → 1101)
Processes (operations)	1. Get n from the user. 2. If n = 0, the answer is "0". 3. Otherwise, repeat: remainder = n MOD 2; n = n DIV 2; put the remainder at the left of the binary digits found so far. 4. Display the binary result.
Loop condition (when the loop stops)	The loop keeps running WHILE n > 0. It stops when n becomes 0 (the last division gives a quotient of 0).
Variables needed	n – integer, the number being divided (changes each pass) remainder – integer, 0 or 1 (n MOD 2) binary – string, the result built from right to left
13 and 25 in binary (computed by hand)	13 = 1101 25 = 11001

Part B – Write the Pseudocode

Box 1. Pseudocode for Problem 1 (Part B)

```
START
  DISPLAY "Enter a whole number (0 or greater): "
  INPUT n
  SET binary = ""
  IF n = 0 THEN
    SET binary = "0"
  ELSE
    WHILE n > 0 DO
      SET remainder = n MOD 2
      SET n = n DIV 2
      SET binary = remainder + binary
    ENDWHILE
  ENDIF
  DISPLAY "Binary: ", binary
END
```

When the input is 0, the loop would never run and binary would stay empty, so the IF n = 0 selection sets binary to “0” and the program displays Binary: 0.

Part C – Draw the Flowchart and Desk Check

Figure 1. Flowchart for Problem 1 (Part C)

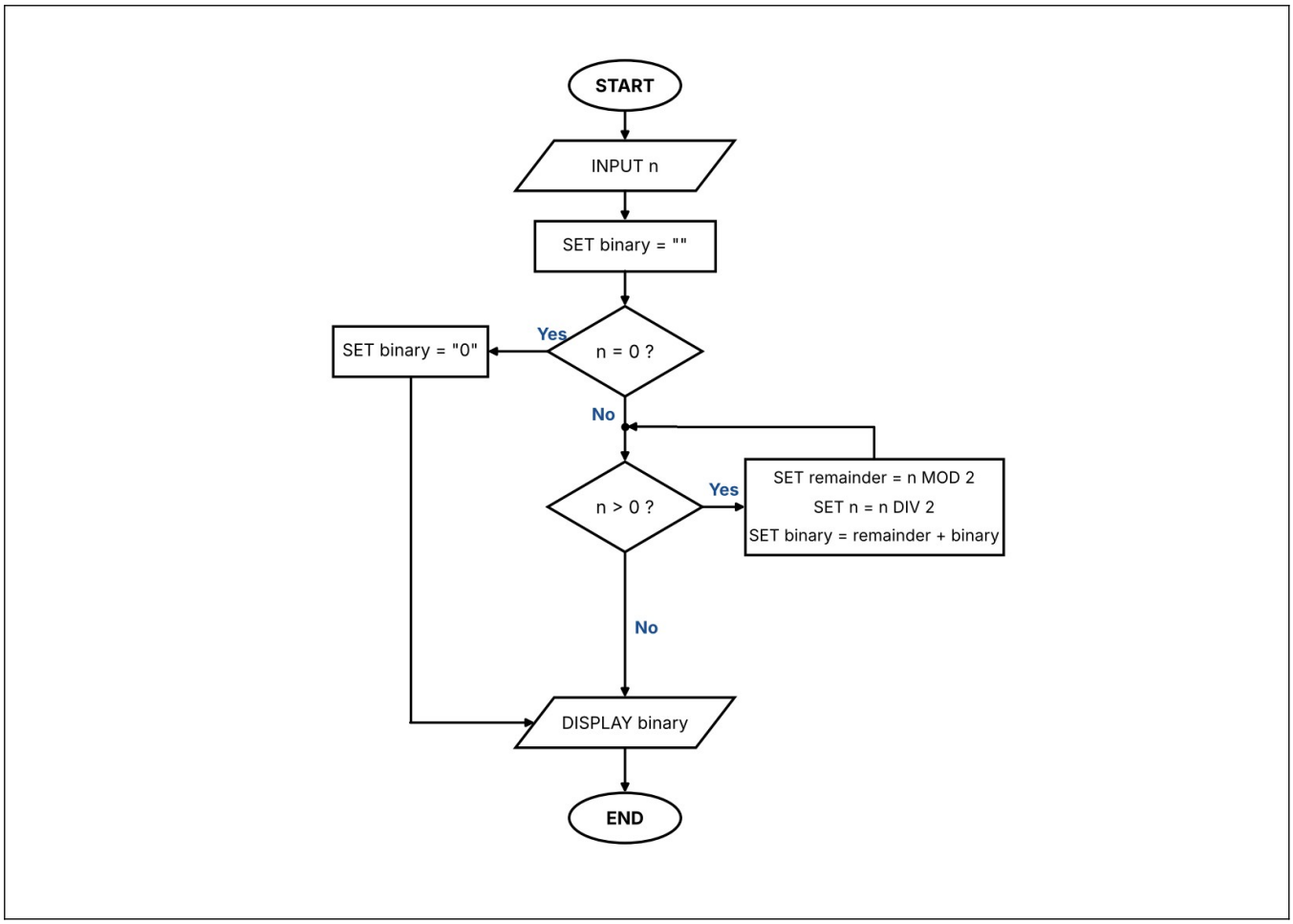


Table 2. Trace of Problem 1 with Input 13 (Part C)

Pass	n (before)	n MOD 2 (remainder)	n DIV 2 (new n)	binary (so far)
1	13	1	6	1
2	6	0	3	01
3	3	1	1	101
4	1	1	0	1101
5	0 (n > 0 is false, loop ends)	—	—	1101

Value displayed at the end: 1101

Table 3. Test Cases for Problem 1 (Part C)

Case	Decimal Input	Binary (by hand)	Binary From Your Pseudocode	Match? (Y/N)
1	0	0	0	Y
2	5	101	101	Y
3	13	1101	1101	Y
4	25	11001	11001	Y
5	255	11111111	11111111	Y

Check: 5 → 5÷2=2 r1, 2÷2=1 r0, 1÷2=0 r1 → 101. 255 → eight remainders of 1 → 11111111.

Part D – Reverse Conversion (Problem 2)

Box 2. Pseudocode for Problem 2 (Part D)

```
START
  DISPLAY "Enter a binary number (digits 0 and 1): "
  INPUT binary
  SET decimal = 0
  SET place = 1
  WHILE binary > 0 DO
    SET digit = binary MOD 10
    SET decimal = decimal + (digit * place)
    SET binary = binary DIV 10
    SET place = place * 2
  ENDWHILE
  DISPLAY "Decimal: ", decimal
END
```

Figure 2. Flowchart for Problem 2 (Part D)

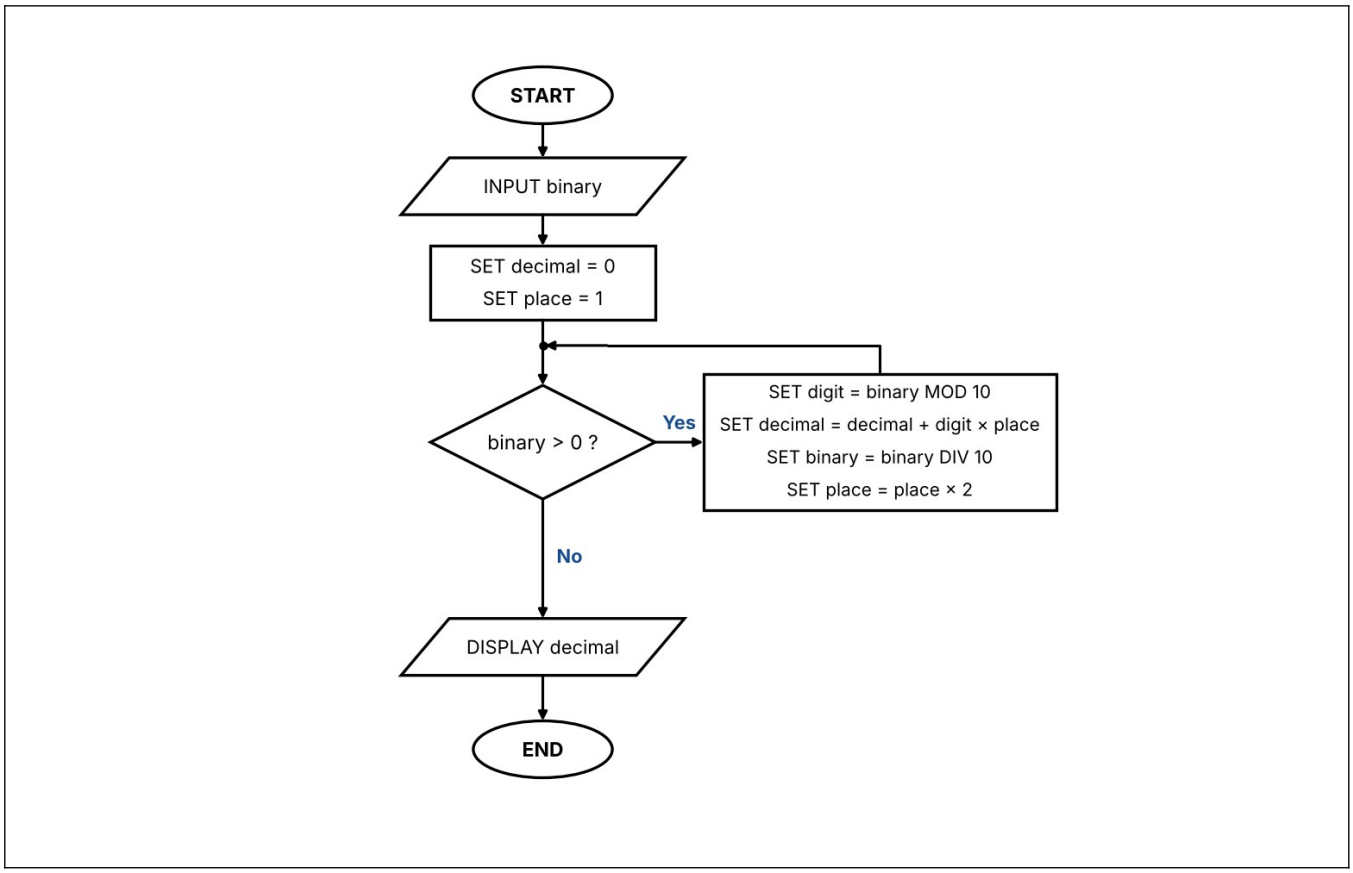


Table 4. Test Cases for Problem 2 (Part D)

Case	Binary Input	Decimal (by hand)	Decimal From Your Pseudocode	Match? (Y/N)
1	1101	1 + 0 + 4 + 8 = 13	13	Y
2	101010	0 + 2 + 0 + 8 + 0 + 32 = 42	42	Y
3	11111111	1+2+4+8+16+32+64+128 = 255	255	Y
4	10000000	1 × 128 = 128 (all other digits are 0)	128	Y

Bonus Box. Extended Pseudocode: Decimal to Base 2, 8, or 16 (Optional)

```
START
  SET digits = "0123456789ABCDEF"
  DISPLAY "Enter a whole number (0 or greater): "
  INPUT n
  DISPLAY "Choose the target base (2, 8, or 16): "
  INPUT base
  IF base is not 2, 8, or 16 THEN
    DISPLAY "Invalid base"
  ELSE
    SET result = ""
    IF n = 0 THEN
      SET result = "0"
    ELSE
      WHILE n > 0 DO
        SET remainder = n MOD base
        SET symbol = digits[remainder]     (10 to 15 become A to F)
        SET result = symbol + result
        SET n = n DIV base
      ENDWHILE
    ENDIF
    DISPLAY "Result: ", result
  ENDIF
END
```

Examples: 13 in base 2 → 1101; 255 in base 8 → 377; 255 in base 16 → FF; 13 in base 16 → D.

Part E – README.md (offline)

Folder layout created in Documents:

```
cc101-portfolio\
  README.md
  week07-algorithms\
  images\
```

README.md content (as uploaded to GitHub):

````
# CC 101 Computing Portfolio

This repository contains my laboratory outputs and documentation for *CC 101
Introduction to Computing* at **Sorsogon State University – Bulan Campus**.

## About Me

- **Name:** Yhamel M. De Guzman
- **Program and Section:** BSCS 1
- **Interests:** Programming, problem solving, and learning new technologies
- **Goal:** To become a skilled software developer who builds useful programs and
keeps learning every day.

## Course Information

| Item | Details |
| --- | --- |
| Course | CC 101 Introduction to Computing |
| Instructor | John Mart D. Gabrentina |
| Academic Year | 2026–2027 |

## Portfolio Contents

1. [Week 7 – Pseudocode and Flowcharts](week07-algorithms/)
2. More activities will be added each week.

## Sample Pseudocode: Decimal to Binary

```
START
  DISPLAY "Enter a whole number (0 or greater): "
  INPUT n
  SET binary = ""
  IF n = 0 THEN
    SET binary = "0"
  ELSE
    WHILE n > 0 DO
      SET remainder = n MOD 2
````
````

```
      SET n = n DIV 2
      SET binary = remainder + binary
    ENDWHILE
  ENDIF
  DISPLAY "Binary: ", binary
END
```

```
## My Workspace

![My laboratory computer](images/my-pc.jpeg)

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*Last updated: 2026-10-20*
```

Table 5. README.md File Check (Part E)

Item	Your Findings
Full folder path of cc101-portfolio	C:\Documents\cc101-portfolio
File name shown in File Explorer	README.md
Type shown in File Explorer (Details view)	Markdown Source File (.md)
Size of README.md shown by dir (bytes)	1,192 bytes
Does type README.md show your Markdown text? (Y/N)	Y
Notepad formatted view available? (Y/N)	Y
Elements that appeared formatted in Notepad (if available)	Headings, bold and italic text, bulleted list, numbered list, table, and link appeared formatted. The code block still looked like plain text.

Part F – Peer Review

Table 6. Peer Review Checklist (Part F)

Reviewer's name: _____ Owner of the README: Yhamel M. De Guzman

Element	Markdown Used	Present and Correct? (Y/N)
Level-1 heading	#	Y
Level-2 headings	##	Y
Paragraph	Text with a blank line before and after	Y
Bold and italic text	**bold** *italic*	Y
Bulleted list	- item	Y
Numbered list	1. item	Y
Table	| --- | header separator	Y
Link	[text](url)	Y
Image	![alt](path)	Y
Code block	Lines between two lines of three backticks	Y
Horizontal rule	---	Y
File saved as README.md (not .txt)	—	Y

The reviewer writes their own name on the line above.

Part G – Take-Home: Upload to GitHub

Table 7. GitHub Submission (Part G, Take-Home)

Item	Your Answer
GitHub username	Yhamel-gu
Repository URL	https://github.com/Yhamel-gu/cc101-portfolio
Date uploaded (YYYY-MM-DD)	2026-10-20
Does the README display formatted on GitHub? (Y/N)	Y
Elements that did not display as expected, and why	At first the Week 7 link and the image did not work. GitHub does not store empty folders, so the link only worked after week07-algorithms had a file in it. The image showed as broken because my photo was my-pc.jpeg while the README said my-pc.jpg; I fixed it by changing the README to images/my-pc.jpeg. Everything now displays correctly.

VI. Realization (Reflection)

1. Which was easier for you to create, the pseudocode or the flowchart? Why?

The pseudocode was easier because it reads like ordinary English steps, so I could write it line by line and just indent the loop. The flowchart took more care, since I had to choose the right symbols, keep the arrows from crossing, and make the loop arrow return to the decision. However, drawing the flowchart helped me see the loop and the Yes/No paths more clearly than the pseudocode did.

2. Did tracing and testing reveal any errors in your pseudocode or flowcharts? Describe what you found and how you corrected it.

Yes. My first draft used SET binary = binary + remainder, and the trace for 13 gave 1011 instead of 1101 because the digits were in reverse order. I fixed it by placing the new remainder on the left (SET binary = remainder + binary). The test case 0 also showed a problem: the loop never ran, so nothing was displayed, which I solved by adding IF n = 0 THEN SET binary = "0".

3. How is writing a README in Markdown different from typing a document in Microsoft Word?

In Markdown I type plain text and add symbols such as #, *, and - to show headings, emphasis, and lists, so I could write it in Notepad without any toolbar. In Word I click buttons and see the final look right away, and the file is saved in a special .docx format. A Markdown file stays small and readable anywhere, and GitHub turns the symbols into a nicely formatted page.

4. How can a GitHub portfolio help you as a future computing professional?

A GitHub portfolio shows my work in one public place, so employers and teachers can see what I can really do. It also keeps a history of my changes and shows how my skills grow from Week 7 onward. Later, it will let me share code and work with other people on projects.